

Kristal Farms Partner Overview

Labrador Coast Hydro / Compute / Heat Project

First target: Nain, Nunatsiavut

Expansion logic: Labrador coastal communities

Document status: Partner discussion draft

Intended audience: Infrastructure partners, hydro/utility partners, compute tenants, telecom/fibre partners, community and government partners

Distribution: External-facing; detailed economics and source archive reserved for later diligence / NDA

1. One-page executive summary

Kristal Farms is a proposed Labrador coast infrastructure model that combines local renewable power, modular compute, fibre connectivity, and useful heat recovery into one community-scale platform.

The project thesis is simple:

Co-locate compute with cold hydro, place modular data containers near village heat users, export compute results by fibre, reuse server heat locally, and operate tenant containers under a black-box model that protects tenant data while creating visible community value.

The first target is **Nain**, with future replication potential across Labrador coastal communities such as Hopedale, Makkovik, Postville, and Rigolet. Nain is the initial focus because it offers the right combination of strategic need, coastal access, local heat demand, diesel-displacement logic, and proximity to potential community-scale hydro development that can be studied further.

Kristal Farms is not a mega-dam project, not a long-distance power-export project, and not a conventional hyperscale data centre campus. It is a **coastal-accessible, community-scale infrastructure model** built around short connections, modular deployment, heat reuse, and phased proof.

What Kristal Farms builds

Kristal Farms develops a village-adjacent compute and heat platform composed of:

- A local hydro / power integration strategy.
- A short medium-voltage connection to a village or port-area substation.
- A fenced modular compute pad yard using standard containerized units.
- Liquid or rear-door heat capture from compute equipment.
- Heat exchangers and a local heat loop serving public buildings, housing clusters, and seasonal greenhouse demand.
- Fibre connectivity and a network operations layer.
- Black-box tenancy contracts that separate host infrastructure operations from tenant data, models, software, and logs.
- A public metrics and governance framework focused on heat, reliability, community benefit, and environmental safeguards.

Why this matters

Most data centre models treat heat as a waste stream and electricity as the product to move over long distances. Kristal Farms reverses that logic. It keeps electricity local, exports compute over fibre, and turns server heat into useful local energy.

For a Labrador coastal community, this can create several forms of value:

- Lower reliance on diesel heat and diesel electricity over time.
- New local infrastructure tied to clean power and fibre.
- Useful heat for schools, clinics, municipal buildings, homes, and greenhouses.
- Skilled local jobs and training pathways.
- A new revenue base from compute pad leases and infrastructure services.
- A replicable coastal model that can be expanded one pad and one community at a time.

Immediate partner opportunity

Kristal Farms is seeking partners to validate and advance the first Labrador coast deployment pathway. The immediate need is not full-scale construction approval. The immediate need is structured alignment around:

1. Nain-first site confirmation.
2. Hydro and grid pre-feasibility.
3. Fibre and logistics feasibility.

4. Community and FPIC process design.
5. Pilot pad technical design.
6. Heat-user mapping and heat-loop feasibility.
7. Commercial structure and partner roles.

The first decision ask is to support a focused pre-feasibility and partner-alignment phase for a Nain pilot, with Labrador coastal replication treated as the second-stage expansion strategy.

2. Why Labrador coast

The Labrador coast presents a distinctive infrastructure opportunity: cold climate, coastal villages, diesel-replacement need, potential local hydro resources, marine logistics, and a strong case for community-scale infrastructure that does not depend on long high-voltage transmission corridors.

Kristal Farms is focused on **coastal Labrador** because the model requires four conditions to work together:

1. **Local or nearby clean power potential.**

The project needs a renewable energy source that can support phased compute growth without depending on distant grid expansion.

2. **Cold-climate cooling advantage.**

Cold air and cold coastal water reduce cooling burden and improve operating efficiency. The preferred design uses non-contact heat exchange and closed loops rather than evaporative cooling.

3. **Useful local heat demand.**

The compute modules must be near buildings, homes, or greenhouse loads that can use the heat. This is why the containers are placed near the village, not at a remote dam site.

4. **Fibre export path.**

The project exports compute results, not bulk electricity. Fibre availability and network resilience are therefore central to site selection.

This approach is intentionally different from inland mega-projects or large power-export concepts. Kristal Farms is not pursuing new mega-dams, long inland transmission corridors, or broad speculative development across the North. The project is focused on sites where modular compute, local hydro, coastal logistics, fibre, and community heat value can combine in a practical sequence.

Labrador coast replication logic

The first deployment should prove a repeatable pattern:

Local clean power + village-adjacent compute pads + fibre + heat reuse + community governance

Once validated in Nain, the same screening logic can be applied to other Labrador coastal communities. Each future site would still require its own hydro assessment, community process, fibre review, logistics plan, permitting pathway, and heat-user mapping.

3. Why Nain first

Nain is the recommended first target because it offers the clearest initial thesis for a Labrador coast pilot.

Nain combines:

- A northern coastal setting.
- Existing community energy challenges.
- Public-building and residential heat demand.
- Marine logistics potential through seasonal coastal access.
- A strong case for diesel displacement.
- A potential local hydro pathway that requires further validation.
- Strategic relevance as a first Nunatsiavut / Labrador coast demonstration site.

The Nain-first approach does not assume that all technical or commercial questions are already resolved. Instead, it creates a focused place to validate the model before expanding the concept across the Labrador coast.

Nain pilot thesis

For Nain, the project would study whether a local hydro resource and village-adjacent modular compute yard can support:

- A short power connection to a village or port-area energy node.
- Modular compute containers sized to available power and useful heat demand.
- A heat loop serving priority buildings first.
- Seasonal greenhouse heat use.
- Fibre connectivity for compute export.
- A black-box tenancy model for external compute tenants.
- Community governance and benefit-sharing structures.

What must be validated

The Nain case should be advanced only through evidence-based gates. Required validation includes:

- Hydro resource and seasonal flow data.
- Winter firm-power profile.
- Environmental constraints and permitting pathway.
- Short MV route and village substation feasibility.
- Fibre availability, redundancy, latency, and service-level expectations.
- Marine logistics and seasonal delivery windows.
- Candidate pad-yard location.
- Priority heat users and building retrofit needs.
- Community consent process and governance structure.
- Pilot economics and tenant interest.

The project should not present Nain as construction-ready until these items are confirmed.

4. What Kristal Farms builds

Kristal Farms is best understood as an infrastructure platform, not a single technology product.

The platform has seven practical components.

4.1 Local clean power integration

The project uses local hydro as the primary power basis. The preferred architecture avoids long high-voltage export lines. Instead, power is moved over a short medium-voltage connection to a village or port-area substation, then distributed to modular compute pads and heat-system equipment.

This structure reduces the need for long transmission corridors, major remote substations, and large up-front infrastructure commitments.

4.2 Village-adjacent compute pad yard

Compute containers are placed near the village heat users, preferably near the port or village edge, not at the remote hydro site. This keeps heat close to public buildings, homes, and greenhouse loads.

The pad yard would include:

- Prepared modular pads.
- Power handoff and metering.
- Cooling and heat-exchange interfaces.
- Fibre ports.
- Physical security.
- Maintenance access.
- Space for phased expansion.

Each pad can be added, removed, upgraded, or replaced as demand and heat capacity evolve.

4.3 Containerized compute modules

The compute modules are standardized container units that can support different tenant configurations. Tenants may bring their own hardware or lease through a project-approved supplier model.

The site host provides the physical interfaces: power, cooling, heat export, fibre, and site security. The tenant controls its own equipment, workloads, software, data, and models.

4.4 Heat capture and local heat loop

Nearly all IT power consumed by servers becomes heat. Kristal Farms treats that heat as a local asset.

The heat system follows a simple dispatch rule:

Reuse → Store → Reject

First, heat is reused in public buildings, homes, domestic hot-water preheat, and greenhouse loads. Second, excess heat can be stored in short-duration thermal tanks. Third, only remaining heat is rejected through approved non-contact cooling systems or dry coolers.

The preferred system uses two sealed circuits:

- An IT cooling loop for the tenant container.
- A building / community heat loop for heat users.

The loops exchange heat through plate heat exchangers. Fluids do not mix.

4.5 Cooling system

The project avoids conventional chiller-heavy data-centre cooling wherever possible. The preferred cooling architecture uses direct liquid cooling or rear-door heat exchangers, closed-loop circulation, and non-contact heat exchange to a cold coastal source or dry-cooler backup.

Cooling design priorities are:

- Low power usage effectiveness.
- Near-zero water consumption from evaporative cooling.
- No contamination between IT fluids, building-loop water, and the environment.
- Environmental discharge safeguards.
- Redundant cooling path for safe curtailment or shutdown.

4.6 Fibre and network operations

Kristal Farms exports compute results by fibre. Fibre is therefore not a secondary utility; it is core infrastructure.

The site requires:

- A network operations node.
- Fibre backhaul to regional connectivity.
- Pad-level fibre interfaces.
- Monitoring of availability and latency.
- Tenant and community traffic separation.
- Operational procedures for outages and maintenance.

The host does not inspect tenant payloads. Network monitoring is limited to infrastructure-level metrics such as link status, bandwidth volume, availability, and latency.

4.7 Governance, scorecard, and audit framework

The project must be measurable and accountable. Public-facing metrics should include energy efficiency, heat delivered, heat utilization, diesel avoided, fibre availability, uptime, local jobs, training hours, and community benefit indicators.

Governance should include community participation, FPIC process design, benefit-sharing mechanisms, and clear distinction between confirmed principles and open design items.

5. Partner opportunity

Kristal Farms requires several partner types. Not every partner is being asked for the same contribution.

5.1 Hydro / utility partner

A hydro or utility partner helps validate and structure the energy basis of the project.

Potential contributions:

- Hydro resource data.
- Power-system expertise.
- Interconnection planning.
- Substation and medium-voltage design.
- Metering and power-quality standards.

- Operating constraints and safety requirements.
- Coordination with regulators and community energy planning.

5.2 Infrastructure investor

An infrastructure investor supports phased capital formation and risk-controlled development.

Potential contributions:

- Pre-feasibility funding.
- Pilot pad financing.
- Project-development capital.
- Infrastructure ownership model.
- Long-term capital for expansion pads.
- Commercial discipline around gate-based investment.

5.3 Compute tenant

A compute tenant provides demand for pad capacity.

Potential contributions:

- Workload requirements.
- Power density targets.
- Security and black-box requirements.
- Fibre and uptime expectations.
- Preferred lease duration.
- Hardware / container specifications.
- Optional attestation or confidential-compute requirements.

5.4 Modular data centre supplier

A modular supplier supports container design, fabrication, commissioning, and maintenance.

Potential contributions:

- Containerized compute design.
- Liquid-cooling or rear-door heat-exchanger systems.
- Power distribution units.
- Fire safety and monitoring.

- Factory acceptance testing.
- Deployment and maintenance support.
- Spare-parts and replacement strategy.

5.5 Telecom / fibre partner

A telecom or fibre partner validates the compute-export pathway.

Potential contributions:

- Fibre route assessment.
- Backhaul design.
- Redundancy options.
- Latency and availability modelling.
- Operations support.
- Service-level framework.
- Local connectivity benefits for community institutions.

5.6 Community / government partner

Community and government partners are central to legitimacy, benefits, permitting, and long-term operating fit.

Potential contributions:

- FPIC and consultation process design.
- Community priorities for heat allocation.
- Candidate public-building heat users.
- Workforce training priorities.
- Land and site-access discussions.
- Benefits agreement structure.
- Environmental and social safeguards.

6. Site-selection logic

Kristal Farms uses a practical screening framework. A site is attractive only if it can support the whole system, not just one component.

Core screening criteria

1. Coastal or practical marine access

Equipment should be deliverable through realistic seasonal logistics. The project should not overpromise year-round marine access.

2. Local clean power potential

The site needs a credible renewable energy source and a feasible short power connection.

3. Village heat demand

There must be priority users close enough to justify a heat loop: clinic, school, municipal buildings, housing clusters, greenhouse site, or other local heat loads.

4. Fibre path

Compute export depends on fibre availability, upgrade potential, redundancy, and service-level expectations.

5. Community alignment

The project must be developed through a legitimate community process, not imposed as a remote infrastructure project.

6. Environmental feasibility

Hydro, cooling, heat rejection, pad siting, marine logistics, and construction impacts must be reviewed through appropriate studies.

7. Modular expansion potential

The site should allow phased pads rather than requiring a large first build.

Exclusion logic

Kristal Farms should exclude sites or concepts that depend on:

- New mega-dams.
- Major inland transmission corridors.
- Unvalidated large reservoir impacts.
- Remote compute pads too far from useful heat users.
- Fibre assumptions that cannot be verified.
- Community benefit models that are not locally governed.
- Economics dependent on speculative revenue claims.

7. Community and heat value

Heat reuse is not an add-on. It is the central difference between Kristal Farms and a conventional data centre.

Heat-first operating principle

The operating principle is:

Public and community heat value first.

Compute output second.

Heat rejection last.

This means that the system is designed around local heat users from the beginning. Compute capacity should grow in phases as useful heat demand, fibre capacity, and community alignment are confirmed.

Priority heat users

Initial heat users should be selected based on public value, proximity, technical feasibility, and reliability needs.

Likely priority users include:

1. Clinic / health facilities.
2. School and education buildings.
3. Municipal or community buildings.
4. Nearby housing clusters.
5. Domestic hot-water preheat.
6. Greenhouse or food-production facilities.
7. Thermal storage tanks.

Seasonal heat strategy

Winter: public buildings and nearby homes take priority.

Shoulder seasons: heat is split between buildings, storage, and greenhouse use.

Summer: greenhouse and storage become the primary useful sinks, with rejection only when useful options are saturated.

Community benefit logic

The community value comes from several channels:

- Useful heat delivered.
 - Diesel avoided.
 - Lower exposure to fuel logistics.
 - Better local air quality from reduced combustion.
 - Jobs and training.
 - Community-benefit agreements.
 - Local procurement where practical.
 - Potential fibre improvements.
 - Transparent public reporting.
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8. Black-box tenancy and data boundaries

Kristal Farms separates physical infrastructure hosting from tenant compute operations.

The host provides:

- Power.
- Cooling interface.
- Heat export interface.
- Fibre connection.
- Pad security.
- Physical metering.
- Environmental and uptime monitoring.

The tenant controls:

- Servers.
- Data.
- Models.
- Software.
- Workloads.

- Application logs.
- Internal telemetry.
- Encryption and access controls.

What the host may monitor

The host may monitor physical infrastructure metrics, including:

- kW and kWh.
- Voltage and power-quality events.
- Cooling flow and temperature differential.
- Heat exported.
- Pad uptime or heartbeat status.
- Bandwidth volume.
- Fibre availability and latency.
- Technical alarms.

What the host does not access

The host does not access:

- Tenant data.
- Tenant models.
- Tenant application logs.
- Tenant code.
- Tenant packet payloads.
- Sensitive network metadata beyond aggregate infrastructure metrics.
- Internal workload content.

Optional hardware attestation or confidential-compute requirements can be added for tenants that require stronger technical assurances.

9. Commercial model at a high level

The commercial model should be presented in categories, not unvalidated projections.

Potential revenue streams include:

- Compute pad leases.

- Power and cooling infrastructure services.
- Heat service agreements.
- Fibre or network service agreements.
- Optional knowledge / Kristals layer for surplus compute, if later validated.

Potential cost categories include:

- Hydro and power interconnection studies.
- Substation and short MV connection.
- Pad-yard preparation.
- Modular containers.
- Cooling and heat-recovery systems.
- Fibre route and network equipment.
- Port logistics and seasonal transport.
- Operations, maintenance, and monitoring.
- Community benefit commitments.
- Environmental and permitting studies.

SLA concept

The SLA framework should address:

- Power availability and power quality.
- Cooling interface conditions.
- Heat export cooperation.
- Fibre availability and latency targets.
- Maintenance windows.
- Emergency curtailment procedures.
- Tenant data-boundary protections.
- Metering and reporting.
- End-of-lease removal and reversibility.

Detailed pricing, returns, revenue projections, and tenant-specific economics should remain in an NDA data room until the pre-feasibility evidence is stronger.

10. Governance and community legitimacy

Kristal Farms should be developed as a community-first project, not merely as infrastructure placed near a community.

Confirmed principles

The partner-facing governance position should include the following confirmed principles:

- FPIC process required.
- Community benefit structure required.
- Heat allocation priorities must be locally informed.
- Public metrics should be reported regularly.
- Environmental safeguards must be measurable.
- Local jobs and training should be part of the model.
- Governance details must be developed with the community and relevant public authorities.

Open design items

Several governance items should remain open until partner and community discussions occur:

- Final committee composition.
- Which committees are advisory vs. decision-making.
- Heat allocation rules during constrained periods.
- Dispute escalation process.
- Community fund mechanics.
- Revenue-sharing structure.
- Annual review process.
- Data-publication scope.

The partner package should be honest about these open items. Presenting unresolved governance structures as finalized would weaken credibility.

11. Roadmap and decision gates

Kristal Farms should advance through decision gates. Each gate should produce evidence before the next stage begins.

Phase 0 — Partner alignment

Purpose: confirm whether the right partners are willing to study the Nain-first opportunity.

Outputs:

- Partner interest confirmed.
- Confidentiality framework if needed.
- Pre-feasibility scope agreed.
- Initial role map completed.

Phase 1 — Site confirmation

Purpose: test whether Nain has a practical pilot site and heat-user base.

Outputs:

- Candidate pad-yard areas.
- Priority heat users.
- High-level heat-loop route.
- Marine logistics assumptions.
- Initial environmental constraints.

Phase 2 — Community / FPIC process design

Purpose: define the legitimate engagement pathway.

Outputs:

- Community process map.
- Decision points.
- Benefit categories.
- Open governance items.
- Engagement calendar.

Phase 3 — Hydro and grid pre-feasibility

Purpose: validate the power basis.

Outputs:

- Hydro resource data plan.
- Seasonal power profile.
- Interconnection concept.
- MV route concept.
- Substation requirements.

Phase 4 — Fibre and logistics design

Purpose: validate compute export and modular delivery.

Outputs:

- Fibre route and service assessment.
- Redundancy options.
- Latency and uptime expectations.
- Sealift and delivery plan.
- Maintenance access plan.

Phase 5 — Pilot pad

Purpose: design the smallest credible working deployment.

Outputs:

- Pilot pad specification.
- Power/cooling/fibre interface specification.
- Tenant profile.
- Commissioning plan.
- Safety and environmental monitoring plan.

Phase 6 — Heat loop and greenhouse integration

Purpose: prove visible community value.

Outputs:

- Public-building heat design.
- Thermal storage sizing.
- Greenhouse heat-use plan.
- Heat-metering system.
- Public dashboard fields.

Phase 7 — Expansion pads

Purpose: expand only after the pilot proves power, fibre, heat, governance, and tenant demand.

Outputs:

- Expansion criteria.
- Additional pad design.
- Additional heat users.
- Updated community benefits.
- Replication screening for other Labrador coastal communities.

12. Immediate decision ask

Kristal Farms is not asking partners to approve a full build at this stage.

The immediate ask is to support a structured **Nain-first pre-feasibility and partner-alignment phase**.

Requested decisions

1. Confirm interest in evaluating a Labrador coast pilot.
2. Identify partner role: power, capital, fibre, compute tenancy, modular supply, community/government support, or technical diligence.
3. Agree on the pre-feasibility scope.
4. Identify data required from each partner.
5. Establish confidentiality rules for sensitive material.
6. Begin site, heat, hydro, fibre, and governance workstreams.
7. Define the evidence required for a pilot investment decision.

What success looks like after the next phase

A successful next phase would produce:

- A validated Nain pilot concept.
 - A clear partner-role map.
 - A preliminary heat-user plan.
 - A hydro and grid evidence plan.
 - A fibre/logistics evidence plan.
 - A community process pathway.
 - A pilot pad technical outline.
 - A risk register.
 - A decision package for whether to proceed to detailed feasibility.
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13. What this package does not claim

To stay credible, the current partner package should avoid overclaiming.

This overview does not claim:

- That the Nain site is construction-ready.
- That hydro output is confirmed without further study.
- That fibre service levels are already contracted.
- That community consent has already been obtained.
- That all governance structures are finalized.
- That year-round marine access is guaranteed.
- That financial returns are proven.
- That all heat can be reused in every season.
- That Kristals or knowledge services are required for the core project to work.

The core project should stand on the physical infrastructure thesis: local clean power, modular compute, fibre export, heat reuse, black-box tenancy, and community value.

14. Closing thesis

Kristal Farms is a Labrador coast infrastructure proposal built around a practical inversion of the conventional data centre model.

Instead of exporting electricity over long distances, the project keeps clean power local and exports compute by fibre. Instead of rejecting server heat, it routes that heat to buildings, homes, storage, and greenhouses. Instead of building a large one-shot campus, it grows by modular pads. Instead of asking the host community to accept a black-box project, it gives compute tenants black-box data protection while making the physical benefits visible through heat, jobs, training, metrics, and governance.

Nain is the first test of that model. Labrador coastal replication is the expansion logic.

The next step is a focused partner-alignment and pre-feasibility phase to determine whether the Nain pilot can be validated technically, commercially, environmentally, and through the required community process.